MIL–P–14574A(MR)
1 MAY 1964
SUPERSEDING
MIL–P–14574(ORD)
30 JUNE 1957

MILITARY SPECIFICATION

PACKING (WIPER RING), SYNTHETIC RUBBER, CLOTH INSERTION

1. SCOPE

1.1 This specification covers one type of synthetic rubber packing used on recoil mechanisms as a wiping ring for the recoiling gun tube.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

CCC–T–191 —Textile Test Methods

MILITARY

MIL–P–14232 P6167049 —Parts, Equipment and Tools for Ordnance Materiel, Packaging of

MIL–P–14232 —P6167049—Packaging Data Sheet for Packing, Wiper Ring

STANDARDS

MILITARY

MIL–STD–105 —Sampling Procedures and Tables for Inspection by Attributes

MIL–STD–417 —Rubber Compositions, Vulcanized General Purpose, Solid (Symbols and Tests)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Bid samples.** The contractor shall submit, with his bid, samples of the rubber compound and of the fabric to be used in fabricating and molding the packing (see 6.1). A certification to the effect that the final packing shall comply with the requirements of this specification shall accompany the samples. Samples of rubber compound shall consist of 3 pads, 6 by 6 by 0.075 inch; and 4 plugs, 1.129 ± 0.005 inches in diameter and 0.500 ± .010 inch thick. A sample of the fabric shall consist of 1 linear yard (see 4.5).

3.2 **Materials.**

3.2.1 *Rubber compound.* The rubber compound shall conform to Grade SC615 of Standard MIL–STD–417.

3.2.2 *Fabric.* The fabric shall be cotton duck, evenly and firmly woven; free from unsightly defects, dirt, knots, holes, weak places, bumps, and irregularities of twist, and consistent in all respects with good manufacturing practice. The fabric shall be suitable for frictioning (impregnation) with synthetic rubber and when tested as specified in 4.5.2 shall conform to the physical properties specified in table I.

FSC 5330

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TABLE I. *Cotton duck*

Weight per square yard (minimum)	Thread Count		Gauge (inches) (min.)	Breaking strength pounds	
	Warp (min.)	Filling (min.)		Warp (min.)	Filling (min.)
10 ounces..	32	28	.031	135	100

3.3 Construction. The packing shall consist of layers of cotton duck fabric impregnated with a rubber copound. Six to 10 plies of fabric shall be uniformly distributed throughout the rubber compound. This construction shall be press cured to the shape specified in Figure 1 (see 6.1 for applicable drawing) so that the compound and fabric become an integral part of each other.

3.4 Coating. The packing shall be given a smooth surface treatment with a high melting point wax or mixture of waxes, and a coating of powdered graphite. There shall be no surface contamination.

3.5 Workmanship. The packing shall be free of roughness mold marks, pits, tears, cracks or breaks, and the presence of foreign particles or grit (see 4.4).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. Unless otherwise specified, a lot shall consist of all packings produced under essentially like conditions and offered for acceptance at one time.

4.3 Sampling.

4.3.1 *Sampling for inspection.* Sampling for inspection of packing for defects listed in table II shall be in accordance with Standard MIL–STD–105.

4.3.2 Sampling for tests: Samples of rubber and fabric that are representative of the lot shall be selected for tests. Quantities shall be as follows:

(a) Rubber: Samples of rubber compound shall consist of 3 pads, 6 by 6 by 0.075 inch, and 4 plugs 1.129 ± 0.005 inches in diameter and $0.500 \pm .010$ inch thick.

(b) Fabric: A sample of the duck shall consist of 1 linear yard.

4.4 Examination. Sample units of packing selected in accordance with 4.3.1 shall be examined for the defects and at the acceptable quality levels shown in table II.

4.4.1 *Examination of fabric.* A sample of the fabric shall be examined for suitability for frictioning in accordance with the requirements of 3.2.2.

4.4.2 *Examination of packaging, packing and marking.* Examination shall be made to ascertain that the packaging, packing and marking of the material conform to this specification.

4.5 Tests. All tests of this specification shall be required on both the rubber compound (see 4.5.1) and the fabric (see 4.5.2) for both the bid sample inspection and acceptance inspection.

4.5.1 *Tests of rubber compound.* Samples of rubber compound selected in accordance with 4.3.2 shall be tested in accordance with the applicable requirements and tests of Standard MIL–STD–417.

4.5.2 *Tests of fabric.* Cotton duck selected as specified in 4.3.2 shall be tested in accordance with table III. Unless otherwise specified the method of test in table III shall be that of Specification CCC–C–191.

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TABLE II. *Classification of defects*

Category	AQL percent	Defect	Method of Inspection
Critical		None defined	
Major 101	1.0	Wrong number of fabric plies (see 3.3)	Visual
Major 102		Plies not uniformly distributed (see 3.3)	Visual
Major 103		Wrong shape (see 3.3) or dimensions (see 6.1)	Measure
Major 104		Compound and fabric not intregal part of each other (see 3.3)	Visual
Major 105	2.5	Improper coating (see 3.4)	Visual
Major 106		Incomplete coating (see 3.4)	Visual
Major 107		Surface contamination (see 3.4)	Visual
Major 108	1.0	Roughness (see 3.5)	Visual
Major 109		Mold marks (see 3.5)	Visual
Major 110		Pits (see 3.5)	Visual
Major 111		Tears (see 3.5)	Visual
Major 112		Cracks or breaks (see 3.5)	Visual
Major 113		Indicated presence of foreign particles or grit (see 3.5)	Visual

TABLE III. *Methods of test for fabric*

Test	Requirement	Test Method	Number of Determinations	Reported as:
Weight	Table I	5041	5	Avg. of the 5 determinations to nearest 0.01 ounce
Thread count:				
Warp	Table I	5050	5	Avg. of the 5 determinations to nearest whole number
Filling	Table I	5050	5	Avg. of the 5 determinations to nearest whole number
Gauge	Table I	Micrometer	5	Avg. of the 5 determinations to nearest .001 inch
Breaking strength:				
Warp	Table I	5100	5	Avg. of the 5 determinations to nearest 1.0 pound
Filling	Table I	5100	5	Avg. of the 5 determinations to nearest 1.0 pound

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Packaging, packing and marking shall be as specified on Packaging Data Sheet MIL–P–14232 P6167049. The level of protection shall be as specified in the contract.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Title, number and date of applicable drawing.

(c) Number of packings (wiper rings) required.

(d) Place to send bid sample and certified report (see 3.1).

(e) Level of packaging and packing required (see 5.1).

6.2 Figure 1 is the same as Drawing B6167049. Figure 1 is to designate shape only (see 6.1(b)).

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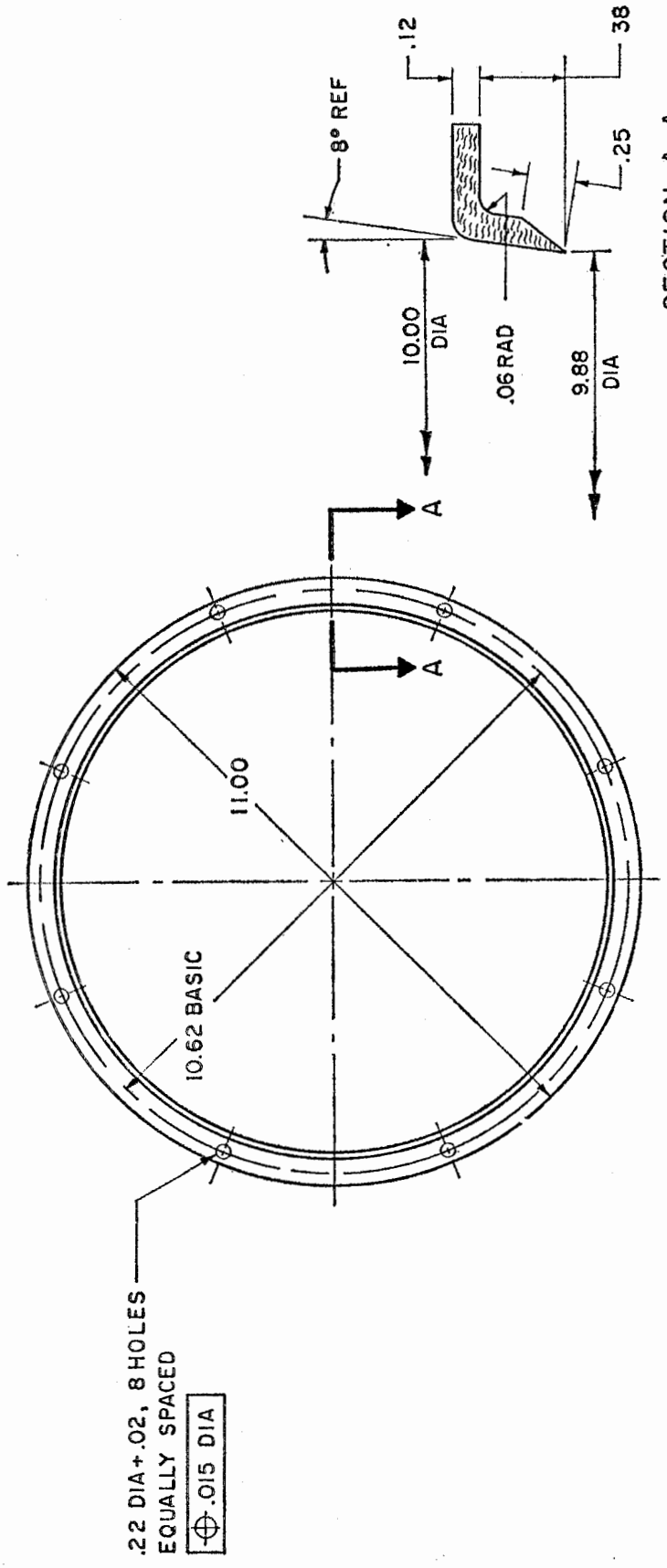


FIGURE 1.

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